

HAT-P-36b

R-0948

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230304 · created 2026-07-29T14:03:47+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460007.8203 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.13 +/- 0.011
Transit depth (Rp/Rs)^2	1.68 +/- 0.28 [%]
Orbital Inclination (inc)	82.94 +/- 2.33
Airmass coefficient 1 (a1)	1.213 +/- 0.015
Airmass coefficient 2 (a2)	-0.169 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.21 %
Best Comparison Star	None
Optimal Aperture	4.4
Optimal Annulus	8.58
Transit Duration (day)	0.0848 +/- 0.0096

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.13 $\pm$ 0.011 vs 0.126 (deviation 0.28 σ)
Duration fit vs published	0.0848 d vs 0.0925 d (deviation 0.62 σ)
Mid-time O-C	12.4 min
Depth SNR	9.1
Residual scatter	1.21 %

Light curve

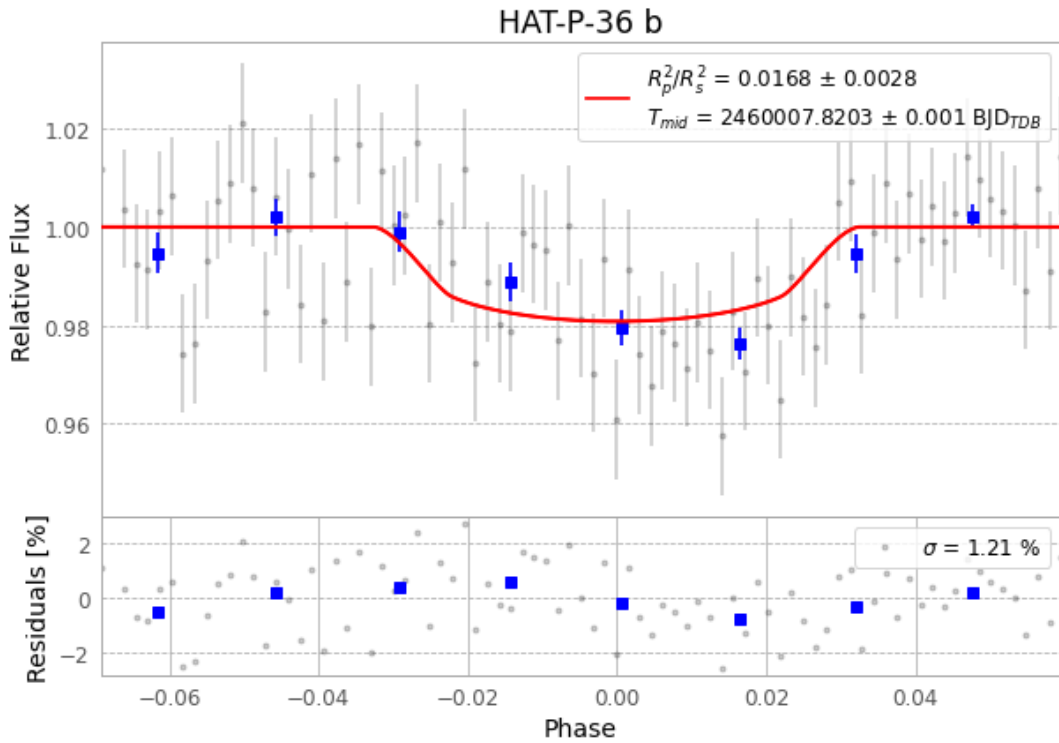


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-03-03.png (SHA-256 7947531fffd6159e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [568, 86], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cd51e38a45e69f70...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

83 FITS frames (55 MB), HATP-36230304052115.FITS → HATP-36230304092715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230304.log (62184f9bd31b...), FinalLightCurve_HAT-P-36 b_2023-03-03.png (7947531fffd6...), FinalLightCurve_HAT-P-36 b_2023-03-03.csv (f107cc41b8ad...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 14:03Z.